DATA ANALYTICS INTERNSHIP

ASSIGNMENT 1

Excel Functions, Pivot Tables & SQL Queries

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Course / Internship: Data Analytics Internship

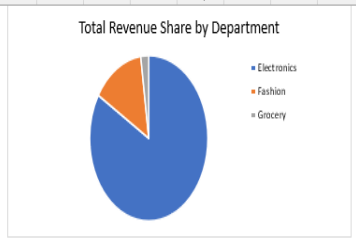
Assignment Number: Assignment 1

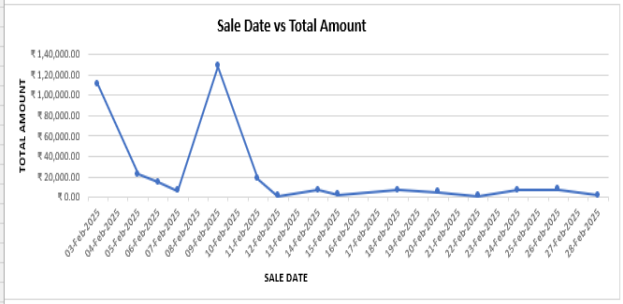
TASK 1: EXCEL FORMATTING, FUNCTIONS & CHARTS

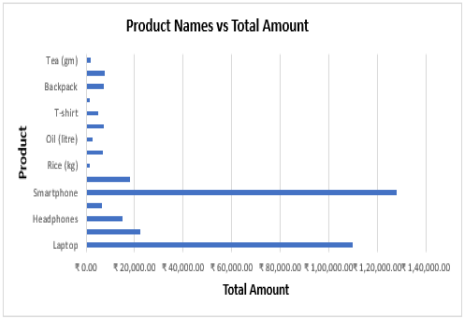
emp_id	name	department	product	quantity	unit_price	sale_date	total_amount	bonus
E01	Aarav	Electronics	Laptop	2	₹55,000	03-Feb-2025	₹1,10,000.00	High
E02	Bhavya	Fashion	Saree	5	₹4,500	05-Feb-2025	₹22,500.00	Normal
E03	Chirag	Electronics	Headphone	10	₹1,500	06-Feb-2025	₹15,000.00	Normal
E04	Diya	Fashion	Shoes	3	₹2,200	07-Feb-2025	₹6,600.00	Normal
E05	Eshan	Electronics	Smartphone	4	₹32,000	09-Feb-2025	₹1,28,000.00	High
E01	Aarav	Electronics	Tablet	1	₹18,000	11-Feb-2025	₹18,000.00	Normal
E06	Farah	Grocery	Rice (kg)	20	₹80	12-Feb-2025	₹1,600.00	Normal
E07	Gaurav	Fashion	Watch	2	₹3,500	14-Feb-2025	₹7,000.00	Normal
E08	Hina	Grocery	Oil (litre)	15	₹180	15-Feb-2025	₹2,700.00	Normal
E03	Chirag	Electronics	Mouse	8	₹899	18-Feb-2025	₹7,192.00	Normal
E02	Bhavya	Fashion	T-shirt	10	₹499	20-Feb-2025	₹4,990.00	Normal
E09	Ishan	Grocery	Sugar (kg)	25	₹55	22-Feb-2025	₹1,375.00	Normal
E04	Diya	Fashion	Backpack	4	₹1,800	24-Feb-2025	₹7,200.00	Normal
E05	Eshan	Electronics	Charger	12	₹650	26-Feb-2025	₹7,800.00	Normal
E10	Jiya	Grocery	Tea (gm)	5	₹350	28-Feb-2025	₹1,750.00	Normal

DEPARTMENTS	TOTAL REVENUE
Electronics	₹2,85,992.00
Fashion	₹48,290.00
Grocery	₹7,425.00

Total_Revenue	₹3,41,707.00
Average_Sales_Valu	₹22,780.47
Max_Sales_Value	₹1,28,000.00
Min_Sales_Value	₹1,375.00
Fashion_Sales_Count	5
Electronics_Revenue	₹ 2,85,992.00







Formulas used:

1. Total Revenue: =SUM(H2:H16)

2. Average Sale: =AVERAGE(H2:H16)

3. Maximum Sale: =MAX(H2:H16)

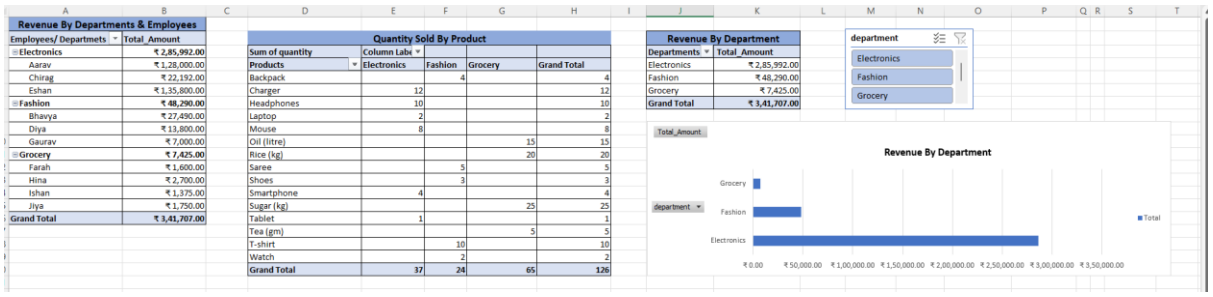
4. Minimum Sale: =MIN(H2:H16)

5. Fashion Sales Count: =COUNTIF(C2:C16,"Fashion")

6. Electronics Revenue
=SUMIF(C2:C16,"Electronics",H2:H16)

7. Bonus: =IF(H2>50000,"High","Normal")

TASK 2: PIVOT TABLES & PIVOT CHARTS



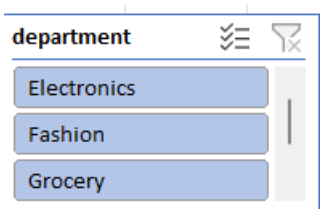
Revenue By Departments & Employees	
Employees/ Departmets	Total_Amount
Electronics	₹ 2,85,992.00
Aarav	₹ 1,28,000.00
Chirag	₹ 22,192.00
Eshan	₹ 1,35,800.00
Fashion	₹ 48,290.00
Bhavya	₹ 27,490.00
Diya	₹ 13,800.00
Gaurav	₹ 7,000.00
Grocery	₹ 7,425.00
Farah	₹ 1,600.00
Hina	₹ 2,700.00
Ishan	₹ 1,375.00
Jiya	₹ 1,750.00
Grand Total	₹ 3,41,707.00

Quantity Sold By Product				
Sum of quantity	Column Labe			
Products	Electronics	Fashion	Grocery	Grand Total
Backpack		4		4
Charger	12			12
Headphones	10			10
Laptop	2			2
Mouse	8			8
Oil (litre)			15	15
Rice (kg)			20	20
Saree		5		5
Shoes		3		3
Smartphone	4			4
Sugar (kg)			25	25
Tablet	1			1
Tea (gm)			5	5
T-shirt		10		10
Watch		2		2
Grand Total	37	24	65	126

Revenue By Department	
Departments	Total_Amount
Electronics	₹ 2,85,992.00
Fashion	₹ 48,290.00
Grocery	₹ 7,425.00
Grand Total	₹ 3,41,707.00

Pivot Table 1 — Revenue by Department & Employees
Pivot Table 2 — Quantity Sold by Product
Pivot Table 3 — Revenue by Department

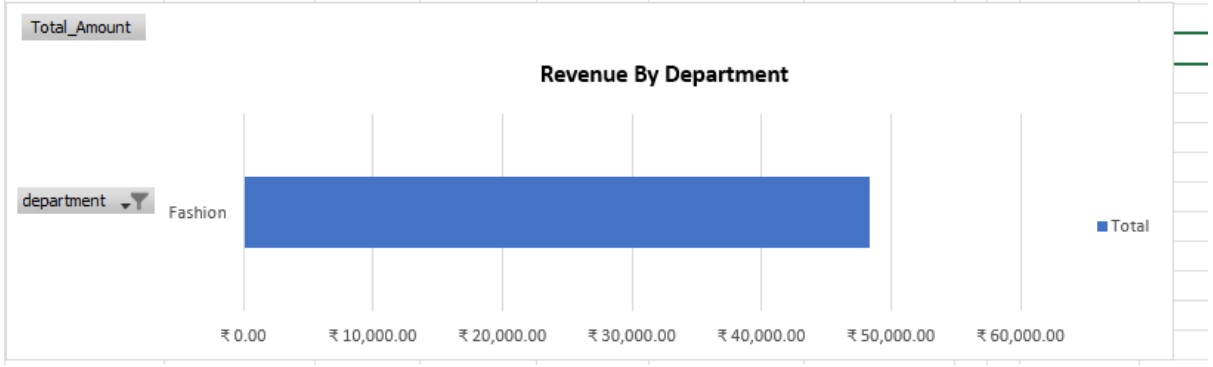
2.2 Department Slicer — Used to filter the Pivot Table and Pivot Chart.



Filter by: Fashion

Revenue By Department	
Departments	Total_Amount
Fashion	₹ 48,290.00
Grand Total	₹ 48,290.00





TASK 3: SQL QUERIES

3.1 SQL Queries

```sql
-- QUERIES
--1. Display all rows of the employees table.
SELECT * FROM employees;

--2. List employees from the 'Fashion' department only.
SELECT name, department
FROM employees
WHERE department = 'Fashion';

--3. Display all employees sorted by salary in descending order.
SELECT name, department
FROM employees
ORDER BY salary DESC;

--4. Display the names of employees whose salary is between 50000 and 70000.
SELECT name, salary
FROM employees
WHERE salary BETWEEN 50000 AND 70000;

--5. Find the total number of employees in each department.
SELECT department, COUNT(name) AS total_number_of_employees
FROM employees
GROUP BY department;

--6. Find the average salary per department.
SELECT department, AVG(salary) AS average_salary_per_department
FROM employees
GROUP BY department;
```

```sql
--7. Find departments where the average salary is greater than 50000 (GROUP BY + HAVING).
SELECT department, AVG(salary) AS average_salary_per_department
FROM employees
GROUP BY department
HAVING AVG(salary) > 50000;

--8. JOIN: Show emp_id, name, product, and amount for all sales.
SELECT e.emp_id, e.name, s.product, s.amount
FROM employees e
JOIN sales s
ON e.emp_id = s.emp_id;

--9. Find the total sales amount made by each employee (JOIN + GROUP BY + SUM).
SELECT e.name, SUM(s.amount) AS total_sales_amount
FROM employees e
JOIN sales s
ON e.emp_id = s.emp_id
GROUP BY e.name ;

--10.Subquery: Find employees whose salary is greater than the average salary of all employees.
SELECT name, salary
FROM employees
WHERE salary > (SELECT AVG(salary)
                FROM employees);
```

OUTPUT OF SQL Queries

```sql
-- QUERIES
--1. Display all rows of the employees table.
SELECT * FROM employees;
```

Data Output | Messages | Notifications

Showing

	emp_id [PK] integer	name character varying (50)	department character varying (30)	salary integer	hire_date date
1	1	Aarav	Electronics	55000	2022-01-10
2	2	Bhavya	Fashion	48000	2021-05-15
3	3	Chirag	Electronics	52000	2023-03-20
4	4	Diya	Fashion	72000	2020-07-01
5	5	Eshan	Electronics	60000	2022-11-12
6	7	Gaurav	Fashion	85000	2019-04-22
7	10	Jiya	Electronics	78000	2020-12-01
8	6	Farah	Grocery	49500	2023-08-05
9	8	Hina	Grocery	46200	2024-01-18
10	9	Ishan	Grocery	60500	2022-09-10

```sql
--2. List employees from the 'Fashion' department only.
SELECT name, department
FROM employees
WHERE department = 'Fashion';
```

Data Output | Messages | Notifications

	name character varying (50)	department character varying (30)
1	Bhavya	Fashion
2	Diya	Fashion
3	Gaurav	Fashion

```sql
--3. Display all employees sorted by salary in descending order.
SELECT name, department
FROM employees
ORDER BY salary DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to

	name character varying (50)	department character varying (30)
1	Gaurav	Fashion
2	Jiya	Electronics
3	Diya	Fashion
4	Ishan	Grocery
5	Eshan	Electronics
6	Aarav	Electronics
7	Chirag	Electronics
8	Farah	Grocery
9	Bhavya	Fashion
10	Hina	Grocery

```sql
--4. Display the names of employees whose salary is between 50000 and 70000.
SELECT name, salary
FROM employees
WHERE salary BETWEEN 50000 AND 70000;
```

Data Output | Messages | Notifications

Showing rows: 1 to 4

	name character varying (50)	salary integer
1	Aarav	55000
2	Chirag	52000
3	Eshan	60000
4	Ishan	60500

```sql
--5. Find the total number of employees in each department.
SELECT department, COUNT(name) AS total_number_of_employees
FROM employees
GROUP BY department;
```

Data Output | Messages | Notifications

	department character varying (30)	total_number_of_employees bigint
1	Electronics	4
2	Grocery	3
3	Fashion	3

```sql
--6. Find the average salary per department.
SELECT department, AVG(salary) AS average_salary_per_department
FROM employees
GROUP BY department;
```

Data Output | Messages | Notifications

Showing rows: 1

	department character varying (30)	average_salary_per_department numeric
1	Electronics	61250.000000000000
2	Grocery	52066.666666666667
3	Fashion	68333.333333333333

```sql
--7. Find departments where the average salary is greater than 50000 (GROUP BY + HAVING).
SELECT department, AVG(salary) AS average_salary_per_department
FROM employees
GROUP BY department
HAVING AVG(salary) > 50000;
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 | Page No:

	department character varying (30)	average_salary_per_department numeric
1	Electronics	61250.000000000000
2	Grocery	52066.666666666667
3	Fashion	68333.333333333333

```sql
--8. JOIN: Show emp_id, name, product, and amount for all sales.
SELECT e.emp_id, e.name, s.product, s.amount
FROM employees e
JOIN sales s
ON e.emp_id = s.emp_id;
```

Data Output | Messages | Notifications

Showing rov

	emp_id integer	name character varying (50)	product character varying (50)	amount integer
1	1	Aarav	Laptop	110000
2	2	Bhavya	Saree	22500
3	3	Chirag	Headphones	15000
4	4	Diya	Shoes	6600
5	5	Eshan	Smartphone	128000
6	1	Aarav	Tablet	18000
7	6	Farah	Rice	1600
8	7	Gaurav	Watch	7000
9	8	Hina	Oil	2700
10	3	Chirag	Mouse	7192

```sql
--9. Find the total sales amount made by each employee (JOIN + GROUP BY + SUM).
SELECT e.name, SUM(s.amount) AS total_sales_amount
FROM employees e
JOIN sales s
ON e.emp_id = s.emp_id
GROUP BY e.name ;
```

Data Output | Messages | Notifications

Showing rows: 1 to 10

	emp_id integer	name character varying (50)	product character varying (50)	amount integer
1	1	Aarav	Laptop	110000
2	2	Bhavya	Saree	22500
3	3	Chirag	Headphones	15000
4	4	Diya	Shoes	6600
5	5	Eshan	Smartphone	128000
6	1	Aarav	Tablet	18000
7	6	Farah	Rice	1600
8	7	Gaurav	Watch	7000
9	8	Hina	Oil	2700
10	3	Chirag	Mouse	7192

```
95  --10.Subquery: Find employees whose salary is greater than the average salary of all employees.
96  SELECT name, salary
97  FROM employees
98  WHERE salary > (SELECT AVG(salary)
99                  FROM employees);
100
```

Data Output | Messages | Notifications

SQL | Showing rows: 1 to 3 | Page No: 1

	name character varying (50)	salary integer
1	Diya	72000
2	Gaurav	85000
3	Jiya	78000

3.2 CRUD Operations

```sql
------- CRUD Operations (Create, Update, Delete)
--1. INSERT: Add a new employee (emp_id=11, name='Kavya', department='Fashion', salary=50000, hire_date='2025-03-01').
INSERT INTO employees (emp_id, name, department, salary, hire_date)
VALUES (11, 'Kavya', 'Fashion', 50000, '2025-03-01');

SELECT * FROM employees;

--2. UPDATE: Increase the salary of all 'Grocery' department employees by 10%.
UPDATE employees
SET salary = salary + (salary * 0.10)
WHERE department = 'Grocery';

--3. DELETE: Remove the employee with emp_id=11 (the one inserted in step 1)
DELETE FROM employees
WHERE emp_id = 11;
```

OUTPUT OF CRUD Operations

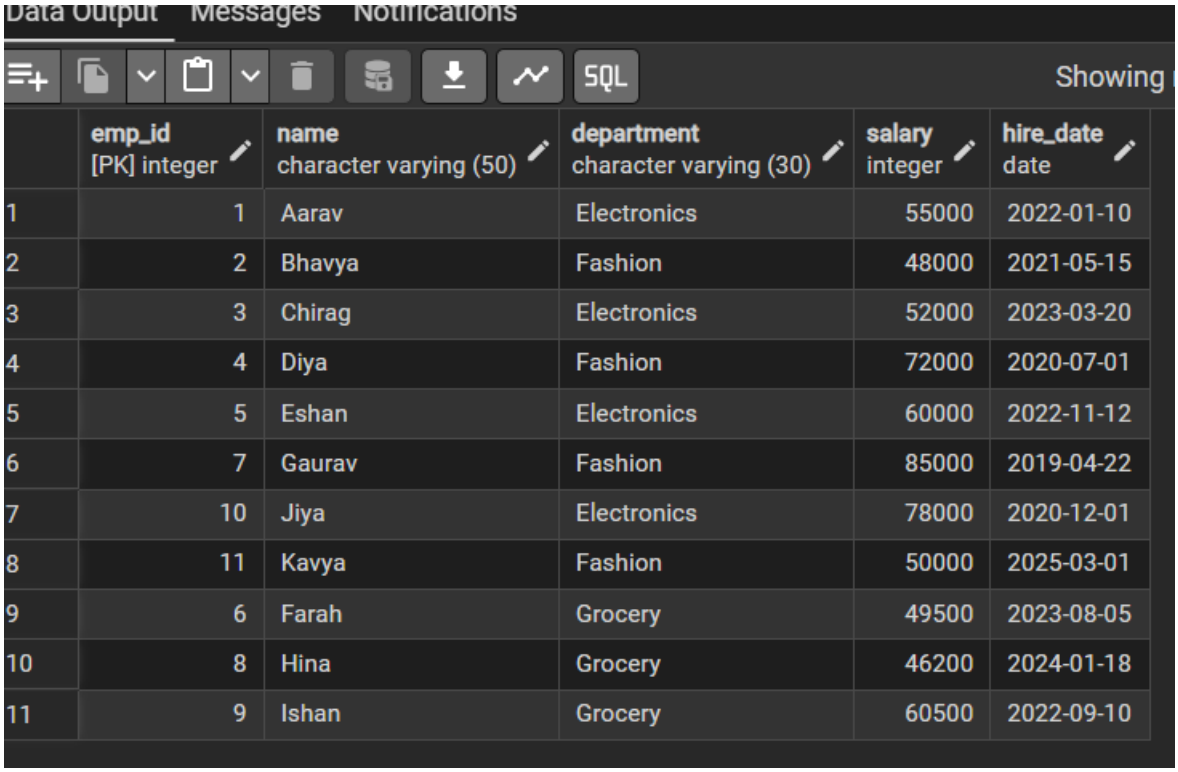
1. INSERT

Data Output | Messages | Notifications

Showin

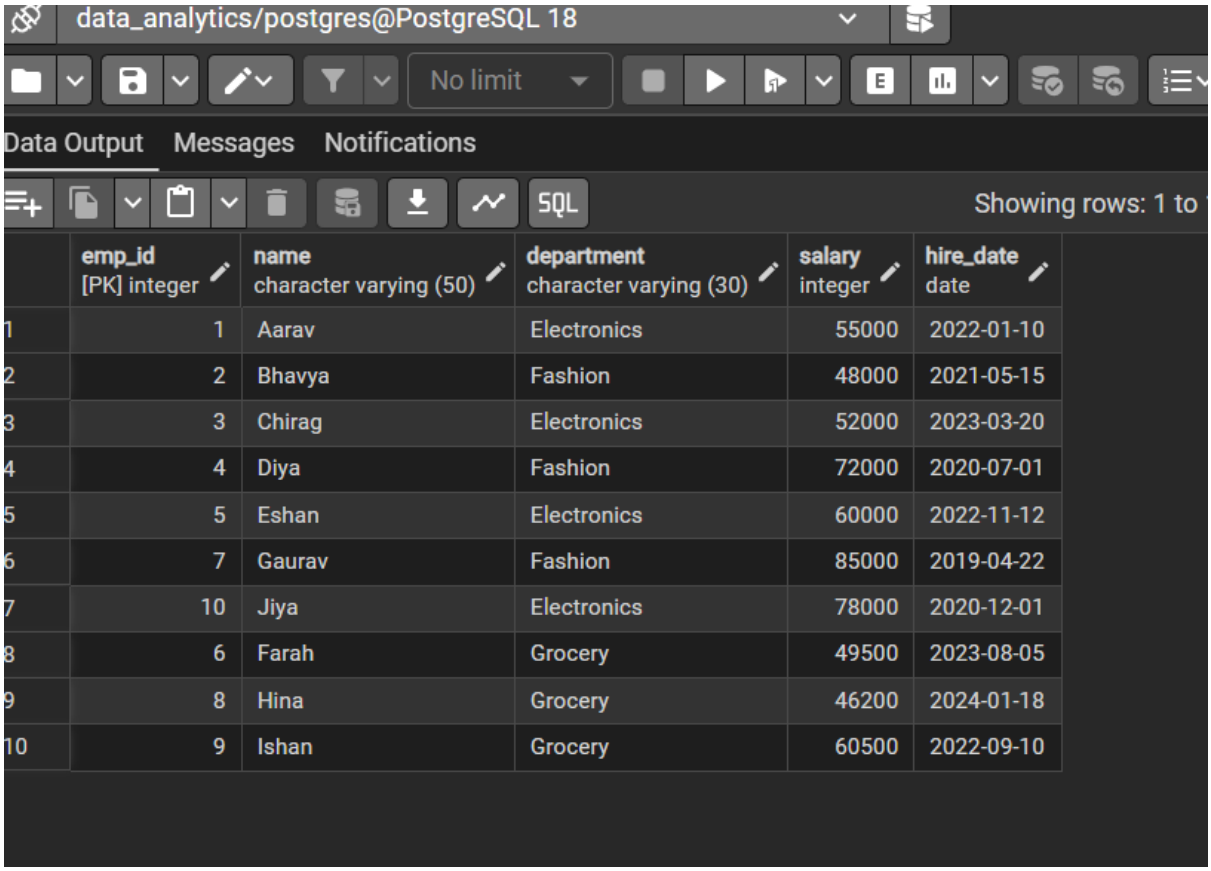
	emp_id [PK] integer	name character varying (50)	department character varying (30)	salary integer	hire_date date
1	1	Aarav	Electronics	55000	2022-01-10
2	2	Bhavya	Fashion	48000	2021-05-15
3	3	Chirag	Electronics	52000	2023-03-20
4	4	Diya	Fashion	72000	2020-07-01
5	5	Eshan	Electronics	60000	2022-11-12
6	6	Farah	Grocery	45000	2023-08-05
7	7	Gaurav	Fashion	85000	2019-04-22
8	8	Hina	Grocery	42000	2024-01-18
9	9	Ishan	Grocery	55000	2022-09-10
10	10	Jiya	Electronics	78000	2020-12-01
11	11	Kavya	Fashion	50000	2025-03-01

2. UPDATE



	emp_id [PK] integer	name character varying (50)	department character varying (30)	salary integer	hire_date date
1	1	Aarav	Electronics	55000	2022-01-10
2	2	Bhavya	Fashion	48000	2021-05-15
3	3	Chirag	Electronics	52000	2023-03-20
4	4	Diya	Fashion	72000	2020-07-01
5	5	Eshan	Electronics	60000	2022-11-12
6	7	Gaurav	Fashion	85000	2019-04-22
7	10	Jiya	Electronics	78000	2020-12-01
8	11	Kavya	Fashion	50000	2025-03-01
9	6	Farah	Grocery	49500	2023-08-05
10	8	Hina	Grocery	46200	2024-01-18
11	9	Ishan	Grocery	60500	2022-09-10

3. DELETE



	emp_id [PK] integer	name character varying (50)	department character varying (30)	salary integer	hire_date date
1	1	Aarav	Electronics	55000	2022-01-10
2	2	Bhavya	Fashion	48000	2021-05-15
3	3	Chirag	Electronics	52000	2023-03-20
4	4	Diya	Fashion	72000	2020-07-01
5	5	Eshan	Electronics	60000	2022-11-12
6	7	Gaurav	Fashion	85000	2019-04-22
7	10	Jiya	Electronics	78000	2020-12-01
8	6	Farah	Grocery	49500	2023-08-05
9	8	Hina	Grocery	46200	2024-01-18
10	9	Ishan	Grocery	60500	2022-09-10